

## FLAGSHIP SOLUTION SCALING TRACK PROFILE

# Precision Nutrient Management at Scale: Supporting Vietnam's One-Million-Ha Low- Emission Rice Programme

Scaling for Impact Science Program | Area of Work 2

**Grand Challenge 2** · Deliver Inputs and Services at Scale through Inclusive Market Systems

**Flagship 2.1** · Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

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# Foreword

## CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

**Flagship 1.1** — Digital Climate Services for Resilient Farm & Enterprise Management

**Flagship 1.2** — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

**Flagship 2.1** — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

**Flagship 2.2** — Entrepreneurship through Mechanization & Postharvest Services

**Flagship 2.3** — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

**Flagship 3.1** — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

# 1. Overview

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Rice production in Vietnam's Mekong River Delta (MRD) is a cornerstone of both national and global food security, yet it faces a critical and growing challenge: the inefficient overuse of nitrogen fertilizers, coupled with substantial nutrient losses from rice straw management. These practices drive high input costs that erode farmer incomes, increase greenhouse gas (GHG) emissions and overall carbon footprint, and contribute to serious water and air pollution.

Our scaling track—**Precision Nutrient Management at Scale (PNM Vietnam)**—directly addresses the said challenges by supporting the Government of Vietnam's ambitious “**One-Million-Hectare High-Quality and Low-Emission Rice Program**” (**1M-Ha Program**), which aims to fundamentally transform rice production systems in the MRD toward greater efficiency, sustainability, and climate resilience.

The core solution being scaled is PNM, anchored by the Rice Crop Manager (RCM) digital decision-support tool. RCM delivers site-specific, field-level nutrient recommendations, building on decades of research in Site-Specific Nutrient Management (SSNM). This core innovation is bundled with complementary technologies—including mechanized direct seeding combined with fertilizer deep placement (mDSR), drone-based fertilizer application, and improved recovery and utilization of rice straw nutrients—to further enhance agronomic efficiency and environmental performance. The scaling pathway prioritizes the integration of Precision Nutrient Management into the 1M-Ha Program's official technical guidelines, the use of public extension systems for large-scale implementation, and strategic engagement with private-sector partners to create strong market-based pull mechanisms that accelerate and sustain farmer adoption.

Extensive evidence justifies this scaling investment. SSNM principles are already validated and incorporated into national recommendations. Pilots across 1,000 hectares in 2024-2025 demonstrated that the bundled innovations can reduce nitrogen fertilizer use by 20-30%, cut pesticide use by 10-30%, and increase income by 5% (from the reduction of input costs). With strong government commitment, a robust partnership network, including government agencies like the Vietnam Ministry of Agriculture and Environment (MAE), National Agricultural Extension Center (NAEC), Vietnam Rice Sector Association (VIETRISA), and private companies, and co-financing from bilateral projects, this track is poised for transformative impact at scale.

## 2. Innovation Bundle and Scaling Ambition

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### 2.1 What is Being Scaled

- **Core Innovation:** The **Rice Crop Manager (RCM)** digital app, which operationalizes SSNM to provide tailored fertilizer recommendations to individual farmers.
- **Bundled Innovations:**
  - **Mechanized Direct Seeding combined with fertilizer deep placement (mDSR):** A system that reduces seed and fertilizer rates and places fertilizer precisely for maximum efficiency.
  - **Drone-based Fertilization:** Ensures precise and uniform application of nutrients, reducing labor and input waste.
  - **Improved recovery and utilization of rice straw nutrients:** mechanized rice straw incorporation combined with microbial inoculants to accelerate in-field decomposition, and circular reuse of rice straw in diversified systems such as rice–potato and rice–sweet potato production.
- **Gender-Responsive Scaling Models:** Approaches that ensure women and youth can equitably participate in and benefit from the new technologies and business models.

## 2.2 Evidence of Impact

- **Pilot Phase Results (2024-2025, ~1,000 ha):** Demonstrated clear benefits, including a **30-50% reduction in seeding rates**, a **20-30% reduction in nitrogen fertilizer use**, and a **10-30% reduction in pesticide use** through mDSR and associated best practices.
- **Proven Principles:** The SSNM principles underlying RCM have been validated over decades and across multiple regions, consistently improving yield, nitrogen use efficiency, and economic benefits.
- **Business Model Viability:** Assessment of mDSR service provider models in Can Tho and Tien Giang provinces showed annual profits of ~5,000 \$US, with a payback period of **1.4 years** on seeder investment.

## 2.3 Scaling Targets

| Outcome Metric                            |  |  | 2026 Target                                          | Long-term Goal (3–5 years)                                             |
|-------------------------------------------|--|--|------------------------------------------------------|------------------------------------------------------------------------|
| Farmers reached                           |  |  |                                                      | 1,000,000 farmers (by 2030)                                            |
| Area under PNM practices (ha)             |  |  | 100,000                                              | 1,000,000 ha in the Mekong River Delta                                 |
| Increase of rice production profitability |  |  | ≥5% (for adopting farmers)                           | ≥5% (for adopting farmers)                                             |
| GHG Emission reduction                    |  |  | Data collection and baseline established             | ≥10% reduction in carbon footprint (LCA)                               |
| Investment leveraged                      |  |  | ~USD 3 million (2026–2028) from WB, USDA, TaiwanICDF | Sustained public and private investment within 1M-Ha program framework |

# 3. Scaling Pathway and Theory of Change

## 3.1 Pathway Strategy

The scaling pathway is built on a **public-private partnership (PPP) model**, embedded within the national 1M-Ha Program, which serves as the primary engine for scale.

- **Institutional Integration:** The core strategy is to embed PNM into the national technical guidelines of the 1M-Ha Program. This shifts PNM from a project-based activity to a routine part of public extension and program implementation, disseminated through the National Agricultural Extension Center (NAEC) and provincial Departments of Agriculture and Environment (DAEs).
- **Private Sector Pull:** Fertilizer companies (e.g., Binh Dien, MTK, CCV, etc.) are engaged by adapting RCM to recommend their specific products, creating a commercial incentive for them to promote the tool and its recommendations to their dealer networks and farmers.
- **Last-Mile Delivery:** Farmer cooperatives, rice producers/ exporters, and service providers (for mechanization, drones, fertilizer dealers, etc.) are the ultimate delivery points. The program focuses on building their capacity and demonstrating the business case for adopting the bundled technologies, moving beyond simple demonstrations to create self-sustaining demand.

### 3.2 Theory of Change

If smallholder farmers and their cooperatives gain access to a proven PNM bundle (RCM + mDSR + rice straw nutrient + digital tools) through the national 1M-Ha Program's extension system and private sector channels, **and** if market and coordination failures are addressed through PPP facilitation (via VIETRISA) and inclusive business models (e.g., women-led services), **then** farmers will adopt more efficient practices, leading to reduced input costs, maintained or increased yields, and lower GHG emissions. This will **contribute to** the long-term systemic goal of a sustainable, low-emission, and highly competitive rice sector in Vietnam.

### 3.3 Key Activities Funded by S4I

- **Technology Integration and Validation:** Support for mapping scale-appropriate technologies, validating the PNM bundle in pilot areas, and developing integrated deployment strategies based on lessons learned.
- **Capacity Building and Scaling:** Conduct Training of Trainers (ToTs), develop technical guidebooks and extension curricula, and support field demonstrations to build capacity within public and private sectors.
- **Inclusive Business Model Development:** Co-design and pilot women-led enterprises and youth engagement networks within the 1M-Ha Program framework.
- **Policy Support and Partnership Facilitation:** Engage with MAE, VIETRISA, and other stakeholders to ensure policy frameworks and incentive mechanisms support PNM implementation.

## 4. Partners and Roles

| Partner Type     | Specific Partners and Quantities                                                                                                                                                                                                                                                                                                               | Role in Scaling Pathway                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Demand Partners  | <b>Farmer Cooperatives and Farmers:</b> ~100,000ha in 2026, scaling to 1M ha in 2030. <b>Exporters and Agribusinesses:</b> Loc Troi, Trung An, and others demanding green and low-emission rice.                                                                                                                                               | <b>Cooperatives:</b> Aggregate demand, coordinate service scheduling, and adopt PNM bundle. <b>Exporters:</b> Create market pull for certified low-emission rice, offering premium prices as an incentive.                                                                                                                                                                                                                                  |
| Scaling Partners | <b>Government:</b> Ministry of Agriculture and Environment (MAE), National Agricultural Extension Center (NAEC), Provincial DAEs (6+). <b>Industry Association:</b> Vietnam Rice Sector Association (VIETRISA). <b>Private Sector:</b> Fertilizer companies (Binh Dien, MTK, CCV), Mechanization companies (Tu Sang), Drone service providers. | <b>MAE/DAEs:</b> Policy alignment, public financing, and integrating PNM into national technical guidelines and extension. <b>NAEC:</b> Leading PPP taskforce, disseminating innovations through ToTs and extension curricula. <b>VIETRISA:</b> Facilitating multi-stakeholder partnerships and PPP models. <b>Private Sector:</b> Providing aligned inputs, services, and promoting PNM recommendations through their commercial channels. |

**4.4 Funding Model:** S4I provides catalytic funding for technology integration, partnership coordination, and capacity building, while the core costs of inputs and services are financed by farmers, government subsidies, and private sector investment within the 1M-Ha program framework.



## 5. Enabling Environment and Risks

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### 5.1 Opportunities

- **Strong Government Ownership:** The 1M-Ha Program is a national priority with dedicated funding, policy support, and a clear implementation framework up to 2030.
- **Proven Policy Alignment:** PNM principles are already recognized in national recommendations (e.g., "1 Must Do, 5 Reductions", recognized technical guideline for 1M-Ha Program), creating a favorable policy pathway for scaling RCM.
- **Existing Multi-Stakeholder Platforms:** The PPP taskforce chaired by NAEC and facilitated by VIETRISA provides a ready-made mechanism for coordination and collaboration.
- **Bilateral Leverage:** Co-financing from major programs (World Bank, USDA, TaiwanICDF) provides substantial resources for field implementation, allowing S4I funds to focus on technical integration and scaling strategy.

### 5.2 Challenges and Mitigation

- **Risk: Insufficient or unstable public financing** for the 1M-Ha Program's delivery mechanisms, slowing provincial-level implementation.
- **Mitigation:** Leverage aligned bilateral projects (USDA FerRight, World Bank MOM-P) to sustain field activities during gaps. Focus CGIAR efforts on technical assistance and capacity building that keeps the pipeline primed for when public budgets rebound.
- **Risk: Profitability and operational complexity of mechanized services** on small, fragmented plots, potentially hindering private sector investment.
- **Mitigation:** Position mDSR as a supporting, not core, innovation with modest scale targets. Conduct detailed business model assessments to identify commercially viable zones and service configurations. Develop the supporting ecosystem (spare parts, financing) in parallel with piloting.
- **Risk: Coordination complexity** for synchronized water management and weed control across numerous smallholders.
- **Mitigation:** Prioritize scaling within farmer cooperatives, which are the most efficient unit for synchronized actions. Use seasonal training with simple decision rules and build coordination capacity over multiple seasons, rather than introducing complex prescriptions all at once.
- **Risk: Displacement of women and youth** if new business models favor existing, male-dominated cooperatives.
- **Mitigation:** Explicitly design and pilot women-led and youth-focused business models in line with the government's social inclusion mandate for the 1M-Ha Program. Dedicated activities with the Women's Union will ensure these groups are not left behind.

## 6. Learning and Adaptation

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### 6.1 Monitoring Focus

- **Process Outcomes:** Are partnerships with NAEC and private sector firms active and effective? Are training materials being adopted by extension services?
- **Behavior Change:** Are farmers changing their fertilizer application rates and methods based on RCM recommendations? Are private companies actively promoting RCM?

- **System Outcomes:** What is the rate of expansion of 1M-Ha program hectares adopting PNM? Are there early signs of institutional ownership (e.g., provincial budgets allocated for PNM training)?

## 6.2 Adaptive Management

- **Quarterly Check-ins:** Formal reviews with the S4I AoW2 team to track progress against milestones, discuss emerging challenges, and adjust work plans.
- **Learning Alliances:** Structured feedback loops with farmers, cooperatives, and extension staff to capture ground-level insights and adapt training and messaging in real-time.
- **Mid-Year Outcome Assessment:** A cost-effective qualitative assessment (e.g., Most Significant Change) in 2027 to document early outcomes and inform the scaling strategy for 2028-2030.

## 7. Next Steps and Timeline (2026)

|    |                                                                                                                                                                                                                  |
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|    |                                                                                                                                                                                                                  |
| Q1 | Finalize 2026 work plan and budget with S4I. Initiate partnership agreements with NAEC and Can Tho DAE. Review and refine RCM adaptation for bundled technologies (mDSR, innovative fertilizers).                |
| Q2 | Conduct ecological and value-chain mapping to identify optimal zones for PNM bundle. Begin development of gender-responsive training materials and business model prototypes. Initiate baseline survey analysis. |
| Q3 | Launch Training of Trainers (ToTs) for extension staff in Can Tho. Establish pilot women-led enterprise models. Conduct field demonstrations and farmer field days across target areas.                          |
| Q4 | Execute endline/behavior change survey for 2026 pilot cohort. Document lessons learned and develop integrated scaling strategy for 2027. Facilitate follow-up meeting with AoW3 on carbon market linkages.       |

**Communications and Knowledge:** Key products will include a series of technical briefs on the PNM bundle, a policy brief for MAE on scaling recommendations, and success stories from women-led enterprises. These will be shared via the CGIAR website, S4I communities of practice, and partner newsletters.

## 8. Contact

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